

Publications

Authors are listed alphabetically by surname, following the convention in the theoretical computer science community.

- [1] **Rectangular Matrix Multiplication in the Low-Bandwidth Model**
Chetan Gupta, Jukka Suomela, and Hossein Vahidi
Preprint. 2026. arXiv:2606.04652. [\[arXiv\]](#)
- [2] **Complexity of computing the anti-Ramsey numbers for paths**
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and Hossein Vahidi
Journal article. Theoretical Computer Science 1044, 2025, article 115271. [\[arXiv\]](#) [\[Journal version\]](#) [\[Conference version\]](#)
- [3] **Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and Hossein Vahidi
Conference paper. SIROCCO 2025, pp. 333–349. [\[arXiv\]](#) [\[Conference version\]](#)
- [4] **Brief Announcement: Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and Hossein Vahidi
Brief announcement. SPAA 2024, pp. 305–307. [arXiv](#) [\[full paper\]](#) [\[Conference version\]](#)
- [5] **Fast Dynamic Programming in Trees in the MPC Model**
Chetan Gupta, Rustam Latypov, Yannic Maus, Shreyas Pai, Simo Särkkä, Jan Studený, Jukka Suomela, Jara Uitto, and Hossein Vahidi
Conference paper. SPAA 2023, pp. 443–453. [\[arXiv\]](#) [\[Conference version\]](#)
- [6] **Approximate Minimum Directed Spanning Trees Under Congestion**
Christoph Lenzen and Hossein Vahidi
Conference paper. SIROCCO 2021, pp. 352–369. [\[Conference version\]](#)
- [7] **Complexity of Computing the Anti-Ramsey Numbers for Paths**
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and Hossein Vahidi
Conference paper. MFCS 2020, pp. 6:1–6:14. [\[arXiv\]](#) [\[Conference version\]](#) [\[Journal version\]](#)